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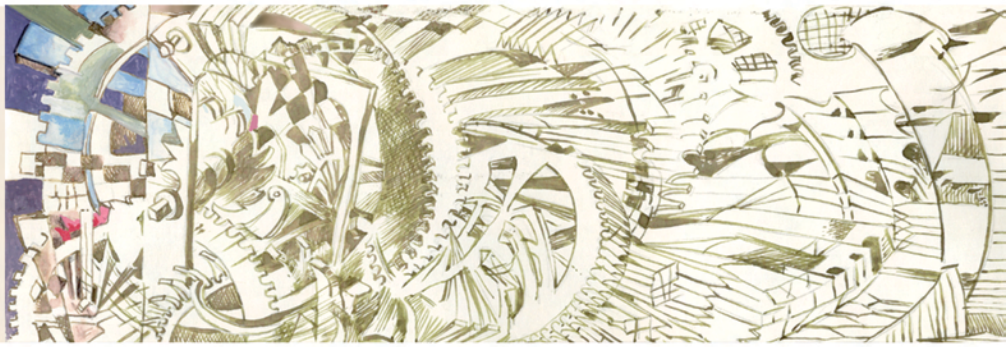
FINANCIAL SERVICES TECHNOLOGY JOURNAL



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*A periodic publication exploring the finer points of topics relevant to the financial services industry*

LETTER FROM THE EXECUTIVE EDITOR

BearingPoint's *Financial Services Technology Journal* brings you views on topics important to decision makers in the financial services industry, presented in a way that puts technology in a true business context.

We believe that **digitization** is one such issue and this issue is entirely dedicated to it.

Surely, the concept of digitization has been around forever: processing data in discrete binary digits is what information technologists and businesses have been doing forever with "0s" and "1s" or their equivalent encoded in vacuum-tube contraptions in times past or, most recently, in atomic quantum states. (There is an hilarious Dilbert cartoon about an old-timer claiming that being able to use "0s" and "1s" is an extravagance available only to the fortunate—and unappreciative—new generations. He goes on saying, "When I started there were no 1s! I once coded an entire database entirely with 0s!")

So what is so special about digitization today? Digitization is the momentous pattern that characterizes what is happening in a wide range of domains in the financial industry: from portal-based claim processing in insurance to straight-through processing in capital markets to integrated commercial loan underwriting in banking. As Richard Walker and Paul Dunay explain in the introductory article "Digitization: The Time is Here," digitization is about actively taking steps to implement an end-to-end solution to support the full value chain, for better controls, standardization, consistency and measurement of the business processes.

The actual underlying disciplines required to make digitization real (such as business process management, content and document management, digital imaging, and workflow and collaboration software) have themselves been around for a while, albeit not until recently with the level of performance or cost to make them either ubiquitous or integrated the way financial institutions are starting to envision now.

And although individual tools supporting these disciplines enjoy double-digit licensing growth, by and large, institutions are implementing these solutions in isolation.

Our point of view here is that achieving the real value these tools offer requires a more encompassing view and the one that we label **digitization** provides such a view, creating critical linkages between the customer and the back office and among the disparate technologies now emerging. Digitization calls for integrated vision and architectures with due focus on domains such as security and enterprise information integration as well as the necessary computing and communications infrastructures.

To explore digitization in a structured way, we have organized the articles using a framework that we prototyped in our inaugural publication and that we continue to use in this issue:

- **The Business of Technology** section explores how technologies create and deliver business value. In "Digitization: The Time is Here," Richard Walker and Paul Dunay describe how digitization can transform financial institutions along the entire value chain of internal and external operations, freeing up financial, human and information resources along the way. They also introduce a "digitization technology stack" that is further discussed in each of the other articles.

The following article brings these concepts together immediately in a very focused manner—In "Enterprise Credit Management: Digitization Comes of Age," Bruce Klein and Sean Coyle describe an approach to address and solve enterprise credit management, to enable financial services firms to bring together all the pieces in the credit management chain. This piece shows how digitization tenets help produce an integrated vision and architecture to create a seamless process that speeds loan closings and significantly improves the efficiency of all personnel involved in the credit management process.

- **How It (Should) All Start** examines the new approaches to strategy and architecture that are shaping the information technology world—In this section, we explore in more detail some of the disciplines that make digitization possible:
  - In "Business Process Management: Taking All the Right Steps," Marc Hurst, Mohammad Karim and Karthik Solaiappan offer a perspective on business process management capabilities, architecture, development and implementation. This perspective is based on actual projects delivered to our customers.
  - In "Document Generation: Digitization in Action," Marcia Douglas discusses a viable approach to digitization that allows it to offer the same benefits as paper-based documents, without incurring the traditional cost of paper publication.
  - In "Enterprise Information Integration: Bridging the Digital Divide," Benedicte Lambert, Punit Kaishap and Chris Rosenberg look closely at a new technology—enterprise information integration (EII). This article focuses on the practical aspects of this technology, including what our customers perceive the role of an EII solution to be, as well as its benefits and challenges.

- **Data For Decision** delves into data and models needed for informed decision making. When we envision digitization, security and compliance are recurring themes that demand utmost attention. Because this domain has grown significantly, we asked Alex Berson, who wrote "Digitization and Security in Financial Services," to help discuss and sort out for you the security issues associated with digitization initiatives in a way that synthesizes what we believe the readership ought to know despite the growing and bewildering range of issues, regulations and technology at play.

The BearingPoint professionals who contribute their insights here are eager to know how well they are doing. We encourage you to offer feedback and ideas for future topics to Pierre Pureur, the *Journal's* editor, and our team at *FSTechJournalEditors@bearingpoint.com*.

**Manuel Barbero**
Executive Editor, BearingPoint's *Financial Services Technology Journal*

# Digitization: The Time is Here

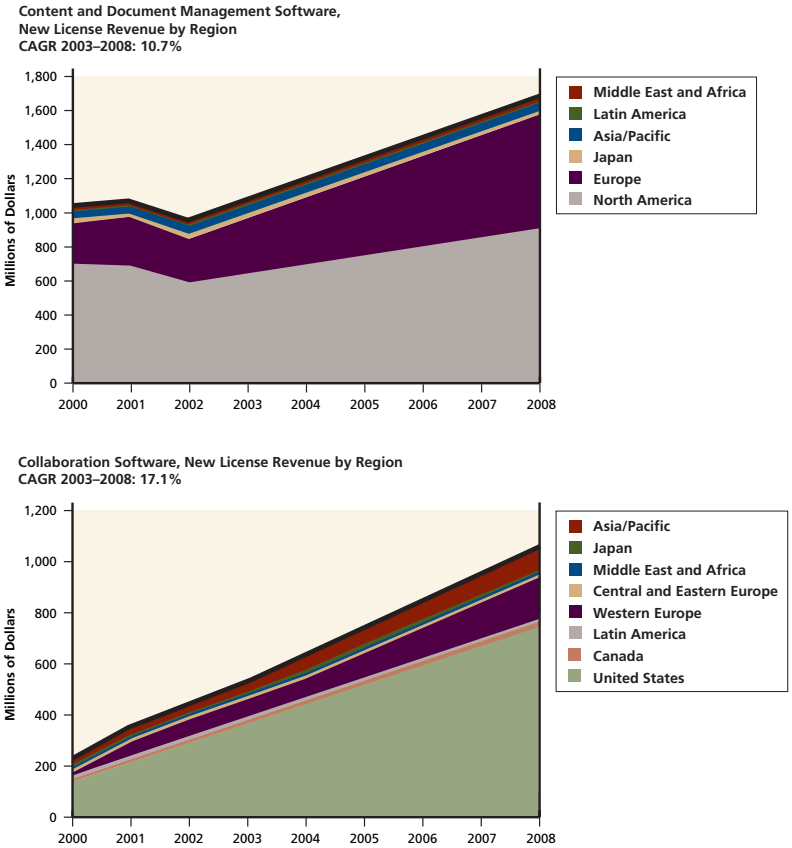
## Abstract

*Financial institutions are taking steps to implement technology in the middle of the value chain, as they recognize the need to bring better controls, standardization, consistency and measurement to business processes and the information that supports them. As a result, tools such as content and document management and collaboration software are forecast to enjoy double-digit licensing growth. Yet by and large, institutions are implementing these solutions in isolation. Achieving the real value these tools offer requires a more encompassing view. Digitization provides such a view, creating critical linkages between the customer and the back office and among the disparate technologies now emerging. Digitization can transform financial institutions along the entire value chain of internal and external operations, freeing up financial, human and information resources along the way.*


From the view of the customer, the financial services industry appears to be embracing digital technology. Each day more and more people sign up for online banking, choosing the convenience of electronic storefronts for paying bills, transferring funds and checking account balances. Some are applying for mortgages and home equity loans online.

The technology march is well underway at the other end of the technology chain, too, in the financial institution back office. As products, customer demands and regulatory requirements proliferate, data warehousing, data convergence and other enterprise tools are helping institutions gain control of the voluminous data they hold on customers and accounts.

**Figure 1: Double-Digit Digitization Growth**




Source: Gartner Dataquest, Forecast: Content and Document Management Software, Worldwide, 2000–2008, Tom Eid, June 17, 2004.

Digital solutions are indeed spreading throughout the financial services industry. Yet often, between the customer and the data stores, paper and manual processes continue to rule. From the knee-to-knee encounter with a customer applying for a loan, to changing beneficiaries on an insurance policy, to arranging artifacts for Know Your Customer (KYC), paper documents handled by human beings remain integral to financial processes. While, in time, more and more of these processes will be automated, there will always be a need for some form of manual intervention.

Of course, financial institutions are taking steps to implement technology in the middle of the value chain. They, like other enterprises, recognize the need to bring better controls, standardization, consistency and measurement to business processes and the information that supports them.

As evidence of this, tools such as content and document management and collaboration software are forecast to enjoy double-digit licensing growth over the next four years (see Figure 1).

Yet by and large, enterprises are implementing these solutions in isolation, both from one another and within the organization. Achieving the real value these tools offer requires a more encompassing view, one that captures the potential of integrating them into a highly cohesive, yet loosely coupled, structure.

Digitization is such a structure. Digitization formalizes, through technology, the business processes, content and associated metadata required to serve customers and support good business decisions. It leverages a service-oriented architecture to create a more nimble business infrastructure that provides critical linkages, both between the customer and the back office and among the disparate technologies now emerging.

## TRAVELING THE ROAD TO THE DIGITIZED ORGANIZATION

The flurry of "e-" activity that began in the late 1990s produced a host of features, functions and business models to capitalize on the explosion in digital technology. The financial services industry joined the trend with online banking and services. Yet as institutions focused on Web strategies on the front end and data access in the back office, they continued to "drop paper" in myriad middle-office manual business processes.

***Digitization formalizes, through technology, the business processes, content and associated metadata required to serve customers and support good business decisions.***

More recently, institutions have begun deploying the new tools emerging for business process management (BPM), workflow, collaboration, and document and content management. Too often, however, these point solutions fail to deliver the hoped-for benefits.

For example, a document management solution can meet the need to store documents in an organized way. But if a team working on a global deal requires distribution of a stored document, the solution does not have the push capabilities to deliver it.

Collaboration software is enjoying robust growth as widely dispersed teams tap it to work together more productively. However, much of this effort is being undertaken in isolation, at the business unit level, similar to the emergence of local area networks a generation ago.

Similarly, a BPM tool can be effective for dealing with highly structured information that all exists within current systems. Unfortunately, businesses operate on unstructured data emerging from person-to-person collaboration, instant messaging, e-mail, and paper and electronic documents. As a result, most business processes are merely a human layer atop a variety of siloed applications that provide the facade of a seamless end-to-end process. Few organizations have actually institutionalized business processes through a software infrastructure, instead relying on people to figure it out.

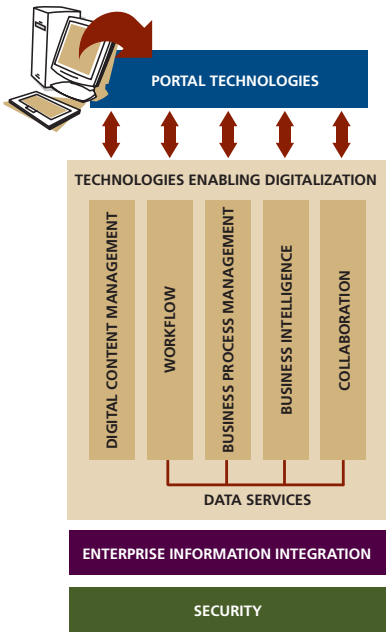
Digitization brings these and other technologies together in a suite—or digitization stack—that captures and manifests the way the institution operates today. At the same time, it supports the ability to dynamically reconfigure systems and information at a moment's notice to implement new business processes in specific contexts, such as a new product or new customer.

A useful construct that can be used to look at how various disciplines interact to deliver the digitization's promise is to look at them as a "stack" of domains that establish the linkage between users (through portals) all the way to foundation technologies and infrastructure tools and techniques.

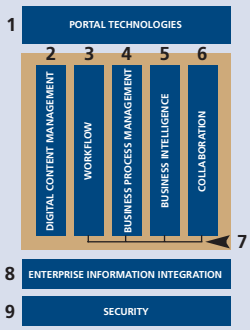
The "digitized stack" (see Figure 2) comprises:

- Portal technologies
- Digital content management
- Workflow
- Business process management
- Business intelligence
- Collaboration
- Data services
- Enterprise information integration
- Security

**Figure 2: The Digitization Stack**




## UNDERSTANDING THE DIGITIZED STACK




**1. Portal technologies** provide a unified work space through which users can access the applications that are deployed throughout the enterprise. The portal can be used as a thin layer on top of existing applications to create a standardized look and feel and aid usability. The enterprise portal also provides the gateway through which the other tools in the digitization stack can be accessed.

**2. Digital content management (DCM)** manages and controls unstructured data, such as video, audio, images, scanned documents and electronic documents. DCM tools provide the ability to create indices and metadata descriptions supporting the search, as well as analysis and easy retrieval of digital content.

**3. Workflow** handles the routing of work between resources, whether those resources are people, systems or machines. It also manages the order in which those steps are handled. At each stage in the workflow, one individual or group is responsible for a specific task. Once the task is complete, the workflow software notifies the individuals responsible for the next task and sends the data they need to execute their stage of the process. It also enables users to reconfigure the flow of a business process following the occurrence of an unforeseen or exceptional event.

**4. Business process management (BPM)** tools model the processes, rules, events and exceptions that exist throughout all, or part of, the enterprise. By integrating with workflow and the other applications in the enterprise, BPM helps manage the processes as they pass between the manual and automated systems.

**5. Business intelligence** tools analyze the vast quantities of data that the enterprise collects and stores and turn it into meaningful information that people can use.

**6. Collaboration** tools expedite information sharing and accelerate decision making between teams and individuals. These tools include online whiteboards, bulletin boards, document collaboration, chat and instant messaging.

**7. Data services** are a key, and often overlooked, part of a service-orientated architecture. They provide a way to centralize the rules governing access to structured data and put the control of providing data to applications back in the hands of the data architect. With data services, application developers no longer need worry about structured query language (SQL) query optimization and the physical location of data, but instead they can concentrate on the business rules and processes.

**8. Enterprise information integration (EII)** tools access the enterprise information, regardless of the source or storage format, and use the existing data resource descriptions to construct a virtual representation that hides the differences of the underlying data sources.

**9. Security** manages and controls the access of all of the users of the enterprise systems. It holds a unified view of the users, their roles, and the different services and functions that they are permitted to access.

Digitization offers the opportunity to use these tools to assemble the true digitized organization, one that can meet the rigors of increasingly competitive markets and growing compliance requirements.

## GATHERING MOMENTUM FOR DIGITIZATION

The lack of progress in the middle of the financial services technology chain is easy to understand. Web applications were relatively low-hanging fruit compared with the messy business of applying business rules to incoming data so you could write it directly to back-end systems, or assembling paper, scanning it into a system, and converting it into dynamic, manageable information.

With the turn-of-the-century collapse of technology company valuations and the slowdown of almost every industry, investment waned. More recently, increased financial markets activity has consumed institutions with servicing accounts, refinancing mortgages, capturing new customers, and coping with a renewed flurry of acquisitions and mergers activity.

Now, however, a confluence of developments is providing the impetus for a renewed digitization push, a crucial step in the creation of the true digital enterprise capable of end-to-end, straight-through processing. Three critical drivers in play make this possible.

First, technologies introduced during the boom years have gone through generations of maturity. Vendors have aggregated, disaggregated and refined technologies to the extent that today a cohesive blend of technologies can be brought together to address processes, features and functions of specific digitization requirements at a reasonable economic investment. Offerings have evolved beyond simple document management or linear process workflow to robust meta solutions for process modeling, document creation, process and document assembly, and manageable multidimensional workflow. Along with these solutions, institutions can apply a metrics infrastructure that provides critical business analytics.

Second, institutions have a strong financial impetus to undertake the transformation. As business has improved, they have chosen to handle transactions by pushing the limits of existing systems and staff, rather than making new investments. The proliferation of many different siloed solutions, with no connecting business process or share data infrastructure, has created unmanageable complexity and business rigidity. Now, to handle still increasing, but unpredictable, volumes, institutions face the choice of increasing head count, with the accompanying impact on margins, or they can rethink, redesign and realign both process and content through digitization.

Finally, government mandates provide the third driver. Know Your Customer, the Sarbanes-Oxley Act, Basel II, the Gramm-Leach-Bliley Act and the Check Clearing for the 21st Century Act, or Check 21, are compelling greater-than-ever access to data, both structured and unstructured, and documents, as well as controls over processes and interactions.

## FACILITATING AND CONTROLLING PROCESS AND WORKFLOW

Digitization manifests in specific ways within various financial services segments. Whether retail banking, insurance or capital markets, underlying the initiative is a discrete set of business processes and content supported by specific technologies that work in a cohesive fashion.

Through digitization, businesses can formalize the business processes they use and manage the structured and unstructured content that goes into or comes out of their systems. They can handle exceptions when there is a deviation to a particular process. And they can improve business intelligence because they will have greater ability to apply metrics to real-time business processes not just derive patterns from data once it has persisted in some back-end system.

Portal and collaboration layers enable people to use the processes and content more effectively. Metrics applied to processes measure their implications for time-to-market, the cost of servicing a product or closing a sale.

Loosely coupled processes will provide the opportunity for disaggregation and re-aggregation of process components into new contexts for new business opportunities. This flexibility promotes the creation of what is variously labeled the agile organization, the zero-latency organization or the real-time enterprise. However it is described, such an organization can distribute a particular task to the optimal economic, geography and demographic locale for execution, using repeatable, measurable, low-cost processes.

Virtually every aspect of the financial services business stands to benefit from digitization.

In mortgage lending, rules-based workflow systems can transform a hand-drafted loan application into a digital stream that arrives at a vice president's desktop for digital approval. The electronic document does not move to underwriting until she applies her electronic signature.

In the retail banking arena, digitization is helping break down intra-institution barriers that have impeded cross-selling at the most ambitious level and the opening of new accounts at the most mundane. By connecting disparate systems with business process automation and management technologies, new accounts can be funded from existing brokerage and asset management accounts. Private banking clients, too, stand to benefit as digitization streamlines the complex and lengthy process for opening such accounts as well as working with aggregated portfolio views with connectivity to real-time market data.

End-to-end processes also reduce the time and cost of deal management. For example, BearingPoint is helping empower an institution to extend its leadership in deal management by helping it set direction to a well-understood, manageable end point. By reducing the physical manifestations of the deal process to a digital content database/repository, the institution will be able to improve deal quality, increase staff productivity and reduce overall portfolio risk.

Finally, digitization will be a crucial link in the end-to-end processes that will support compliance with regulatory requirements. And, as a key component of a complete, traceable document-versioning framework, digitization will further help institutions address legal challenges and fulfill their fiduciary responsibilities.

## DIGITIZATION'S TIME HAS COME

Spurred by business imperatives, legal mandates and technological progress, digitization is moving into the forefront of technology development in financial services. The author views digitization as a vital link between customer experience management, the approach to increasing financial services customer loyalty, and data convergence, through which institutions can halt wasteful technology spending and duplication of effort.

The financial services industry will continue to enjoy booms and endure downturns. Through them, digitization will help institutions achieve the efficiency, performance and customer satisfaction required to succeed in a dynamic, highly competitive marketplace.

## ABOUT THE AUTHORS

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*The Business of Technology*

# Enterprise Credit Management: Digitization Comes of Age

## Abstract

*The time is here for digitization. A potent example of how it can transform financial services is in credit management. For years, financial institutions have leveraged technology to manage the deal/origination pipeline, as well as for booking and servicing closed deals. But for the critical process of internal credit adjudication, the technology play has been more elusive. Now, with an emerging technology platform that the authors refer to as enterprise credit management, financial services firms bring together all three links in the credit management chain, creating a seamless process that speeds loan closings and significantly improves the efficiency of all personnel involved in the credit management process.*


## THE LAST FRONTIER IN CREDIT MANAGEMENT

Financial institutions that extend credit to commercial clients do so through a variety of lending programs. These programs may be differentiated by deal size—for example, small business, middle market, upper middle market and large corporate—or by industry grouping, such as healthcare finance, energy finance, real estate finance, equipment finance and trade finance.

Each of these lending programs has similar operational needs (see Figure 1). They must be able to:

- Track their deal/origination pipeline.
- Manage their internal credit adjudication process.
- Book and service their closed deals in a loan accounting system.

Given that these needs are common across lending programs, financial institutions have attempted to put in place common technology solutions and processes to support these activities. This has met with some success for the first and third activities, where off-the-shelf packages exist and can be tailored to meet specific requirements.

It is the second activity, the management of the internal credit adjudication process, that has eluded a common solution. In many cases, this process is both unique to a particular group and, or perhaps because, it is mostly manual.

This article examines the reasons why the internal credit adjudication process has been so difficult to standardize, and it offers examples of how leading lenders are beginning to address this issue successfully for the first time through digitization.

## DEAL/ORIGINATION PIPELINE MANAGEMENT

Banks sell credit in basically the same way that other industries sell their products or services. They market the uniqueness and breadth of their offerings to attract customers, and then they generally still need to compete with peer institutions on price.

Over the last few years, it has become clear to both financial institutions and software vendors that the sales process for a loan is not so different from that of other, more tangible products. Sales people—called loan originators or relationship managers—sell credit. Customers, or borrowers, purchase loans for a price—that is, index rate plus spread.

**Figure 1: Lending Programs Address Three Common Operations Needs**

In recent years, financial institutions have extended classic customer relationship management (CRM) systems, such as PeopleSoft, Siebel and the Internet-based SalesForce.com, among others, to function as loan/origination pipeline management systems. These systems track the key terms for deals that the bank is pursuing and allow reports to be generated that show management what volume of deals is in the bank's origination pipeline. They also allow relationship managers to track deals and bank managers to calculate sales credit due to relationship managers, whose compensation is generally structured as base salary plus commission on the deals they originate.

A loan/origination pipeline management solution covers the early life cycle of a deal, from initial origination until the time that the bank receives notice from the borrower that it is the chosen lender or has been passed over in favor of another institution.

## BOOKING AND SERVICING

An institution that extends credit may or may not hold that credit to maturity. As long as it is held, though, the bank must book and account for it in such a way that allows for ongoing loan administration and servicing, as well as proper general ledger account posting.

Many commercial lenders use a class of applications known as loan accounting systems to perform these functions. Loan accounting systems can perform servicing functions and act as subledgers with a detailed chart of accounts that enables the lender to post the appropriate debits and credits for activity that occurs within the deal life cycle. These activities include interest rate resets for floating rate loans, draw downs, payments and receipt of fees, among others.

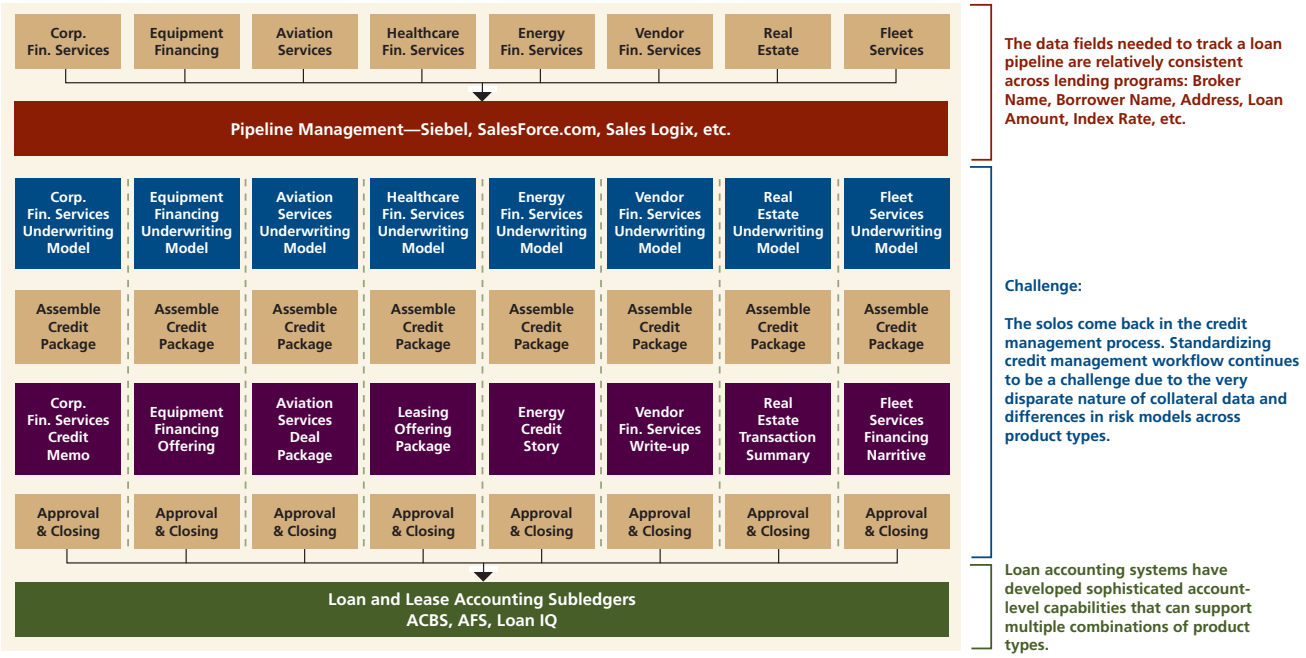
The business, finance and operations managers of a lending program use the reports generated by the subledger system to perform reconciliations, manage risk and gain an overall sense of how the business is performing. At the end of each business day, the system generally rolls up subledger posting activity to a higher level chart of accounts that mirrors the bank's actual general ledger structure. It also passes a daily feed from subledger to general ledger.

Three major vendors of loan accounting systems rule the market today:

- Automated Financial Systems (AFS)
- Fidelity Information Services' Advanced Commercial Banking System (ACBS)
- Misys Loan IQ

These systems all date back to the mainframe era and have benefited from many years of fine-tuning and enhancements. They more than adequately handle loan accounting for all of a wholesale bank's mainstream commercial credit activity.

**Figure 2: The Credit Management Challenge**



Many institutions use more than one of these packages, mainly because of mergers, acquisitions and other legacy events. Given this environment, it is fair to say that standardization of the booking activity on a single technology platform and for an entire suite of wholesale credit products is achievable.

## THE CHALLENGE

In spite of the aforementioned advances in lending support technology, one area of the commercial credit process has until now defied standardization and has traditionally operated as a siloed function in virtually all lending programs: the credit management process.

Credit management refers to the middle tier of the loan life cycle, following origination and preceding closing and booking. It encompasses activities such as cash-flow analysis, underwriting, collateral review and internal credit adjudication. As opposed to deal/origination pipeline management and loan booking and servicing—processes in which loan officers can look at a deal abstractly as a set of key terms—the credit management process represents the point in the loan life cycle in which the data, process and documentation differences between lending programs are most apparent (see Figure 2).

For example, credit decisions for loans secured by commercial real estate are generally based on very different factors than those for unsecured corporate lines of credit. In the former case, the cash flow used to pay back the loan is the cash flow generated by the property. So documents related to the property, such as leases, rent rolls, engineering reports and environmental reports, need to be part of the overall credit package. In the case of an unsecured corporate line of credit, there is no collateral to review and the credit decision is based either on the company's public debt ratings issued by one or more of the rating agencies or by a review of the company's financial statements.

These documentation differences are the reason that many banks' commercial credit packages are a completely paper-based hodgepodge of different file types—for example, Microsoft® Word® and Excel®, Adobe® Acrobat®, GIF and many others—held together with a large binder clip and with an approval signature page on top.

A package of this nature generally does not have consistent page numbering or a table of contents, making it difficult to locate a needed piece of information or to reuse it as a template for a new credit offering. In fact, the original electronic files that made up the credit package may very well reside on someone's personal computer hard drive, making

**Figure 3: Deficiencies of the Current State**

| Current State<br>Credit Management Deficiencies | Current State<br>Cost and Efficiency Implications |
|---|---|
| No centralized document repository for credit documents makes reuse of credit package components on a future deal difficult. | **Increased cycle time**—New offerings start from scratch and components of earlier deals are not retrievable in a standardized, repeatable fashion. |
| • Credit management has limited view of what offerings are close to being submitted for credit review.<br>• Credit package consists of a compilation of files of different types.<br>• Completeness of credit package cannot be verified before submission. | • **Resource management**—Workload balancing within credit group is difficult.<br>• **Slower review process**—Different file types and a nonstandardized credit package make it difficult for credit group to review offerings from different lending programs speedily.<br>• **Incomplete credit packages**—Credit packages may be received that are incomplete, and credit group must iterate with origination until all required data has been submitted. |
| Origination and underwriting unable to see where in the credit process the deal is, who has approved, who still needs to approve, etc. | **Inefficient process**—Origination needs to proactively 'ping' credit management group to ascertain deal status and to be able to report back to borrower. |
| No audit trail exists to track history of the credit process and modifications that may have been made to deal terms due to credit input. | **Regulatory noncompliance**—Organization cannot demonstrate to auditors/regulators what the history of the credit approval was. It cannot be Basel II compliant without this. |
| Loan operations groups unable to see level of activity they should be expecting in the near term. | **Profit and loss reporting issues**—Without a view into the credit pipeline, operations group unable to plan ahead for periods of high activity and may not be able to complete loan bookings on time. |
| A full view is often not available for other credits to same borrower or relationship across the institution. | **Exposure risk**—Inability to view credit relationship with obligor across all lending programs may allow credit risk to approve deal even though total exposure triggers should have kicked in. |
| • No holistic view exists of all credit offerings being worked on prior to credit approval.<br>• Concentration indicators cannot be viewed in aggregate of overall existing portfolio until after approval. | **Concentration risk**—Portfolio may become over-concentrated in certain geography, industry, etc., based on deals currently in pipeline but there is no way for credit group to become aware of this until after approval, when it is too late. |

an institutional history of this credit application virtually impossible.

Adding to the difficulty of standardizing the credit management process are myriad compliance rules defined by an institution's credit risk function. These rules are dynamic and can vary greatly across lending programs within an institution, making them difficult to incorporate within a systematic workflow.

The factors that make the credit management process so difficult to standardize are pervasive, as are the cost and efficiency implications inherent to the silo-based credit management activities (see Figure 3).

## ENTERPRISE CREDIT MANAGEMENT—AN EMERGING SOLUTION

Leading financial institutions today are beginning to solve the credit management challenge by implementing a technology platform that we refer to as enterprise credit management. Enterprise credit management is the implementation of a common process and technology platform to support an entire commercial product suite across the credit life cycle.

Such a solution uses a combination of business process management and content management tools to create a technology platform that adheres to the following guiding principles:

- Store all credit documentation related to a deal electronically and link those credit documents with the deal term data, thereby creating an electronic credit file.
- Automatically generate as much of the credit offering as possible using data already being keyed into a system.
- Standardize a cover page or executive summary for an offering that can work across lending programs and have the enterprise credit management system generate this cover page for each offering.
- Use electronic workflow to move a deal through the credit approval process.
- Provide dynamic dashboards that allow all authorized parties to view the status of a credit offering in real time.
- Automate approval routing to the appropriate authority level based on credit policy.
- Create and maintain an approval log that captures the history of the approval process.
- Have the enterprise credit management system interface directly with booking systems so data does not need to be re-keyed.
- Dynamically generate rules-based compliance worksheets that are filled out online and allow for exception reporting to be easily generated.

An enterprise credit management solution recognizes the process and documentation differences between lending programs and integrates two classes of technologies to deal with this challenge—a business process workflow tool to manage the process differences and a content management tool to manage the documentation differences.

### *Electronic Workflow*

Workflow tools analyze the different data points that a user enters into a system and route deal documents along different paths through the system based on a variety of criteria. For example, based on deal size, a workflow tool can be coded to route the deal to a larger number or smaller number of approvers. Alternatively, based on facility type, a workflow tool can ensure that appropriate extra screens are displayed for users to capture data unique to that facility type.

Using electronic workflow, all offerings can go down the same "assembly line" with the assembly line being smart enough to know which offerings need to stop at which "stations" for data or documents to be added.

Once electronic workflow is used to manage the credit process, what has traditionally been a "black box" process now becomes a transparent and accountable process. No longer must the relationship manager wonder where in the credit department a deal is languishing. Nor must the operations manager be the stepchild of the lending process and live in fear of a flurry of deals arriving for booking when staffing is at its lowest. Real-time dashboards allow a view into the credit process such that, for a particular deal, all stakeholders can see if it has been submitted for approval, who has approved it, who has yet to approve it and what the targeted approval date is—among many other activities.

### *Content Management*

A content management tool creates a common, overall structure that defines what a credit package looks like at a particular institution—that is, which sections must appear, which sections may appear optionally and in what order sections should appear. Different lines of business may require different documents, so placeholders are left within the document structure that can be filled with business-specific content.

Think of the document structure as a series of buckets, some of which must be filled, some of which are optional, some of which the system populates and some of which are uploaded as Word or Excel files or any other type of document to users' hard drives (see Figure 4).

The power of having these documents in a content management tool is that

version control is now possible. If multiple versions of underwriting are performed for a given deal, the content management engine allows earlier versions to be brought back for review, as opposed to a regular file system in which the final version wipes out all previous versions.

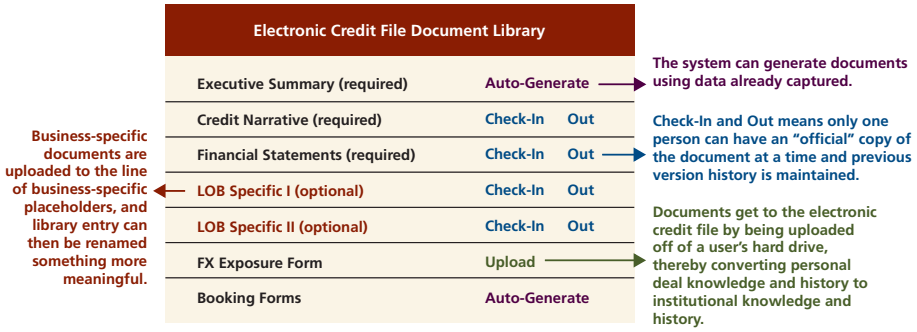
## DOCUMENT ASSEMBLY

The document assembly engine interacts with the content management system (document library) to assemble all the disparate parts contained within into a single portable document format (PDF) document. Want to review the financials? Instead of paging through a 100-page paper document, just click the hot-linked table of contents.

Figure 5 illustrates the three major processes of the document assembly subsystem:

- Template-based PDF conversion
- Uploaded document conversion to PDF
- Final assembly

**Figure 4: Document Structure**




For a more complete discussion of the various XML-related technologies and the overall design patterns used, see "Document Generation: Digitization in Action" in this issue of the *Financial Services Technology Journal*.

## BUSINESS IMPLICATIONS

The business implications of an electronic and structured credit management process are enormous. The "front office" always wants to do more volume but cannot throttle up the origination engine if the risks cannot be understood or the accounting cannot keep up.

With enterprise credit management in place, a credit risk area has a real-time handle on all of the proposed credit activity the institution is contemplating, in the context of current total exposure, making it more comfortable that an increase in volume will not adversely affect its ability to control credit risk. Also, since the system prevalidates much of the package for accuracy and completeness, credit risk personnel need to spend less time chasing down data and documents, thereby allowing more volume to be handled with the same amount of staff.

Similarly, operations personnel can finally "come out of the back office," because they have, for the first time, a complete view of all in-progress credit activity that they can plan and staff for, rather than having the first word of a deal being when the originator drops the closed deal on their desk for booking.

A summary of the many business and process efficiencies gained via enterprise credit management appears in Figure 6.

## TECHNICAL OVERVIEW

The enterprise credit management system uses an application architecture based on the standard n-tier architecture, consisting of a presentation tier, business tier and persistence tier. The system is modeled based on Core Java 2 Platform Enterprise Edition (J2EE) design patterns, which we will discuss later.

**Figure 5: Document Assembly Subsystem Processes**

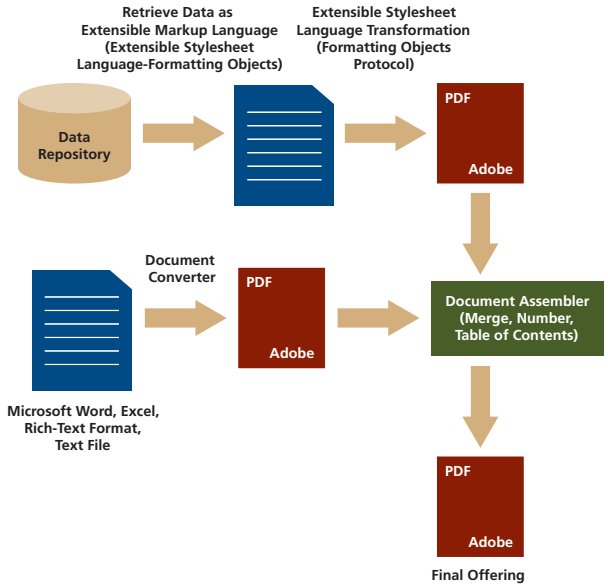


**Figure 6: Efficiencies Gained**

| Efficiencies Gained | |
|---|---|
| 1. | **Dashboards** allow view of credits in progress:<br>• Originators can see the status of deals they have submitted for review.<br>• Credit risk can see what deals are waiting on their approval and what deals are coming down the pipeline and can research deal decisions in the past.<br>• Loan administrators receive all information needed to book the deal and can also see what deals are almost ready to be booked to be able to proactively plan for periods of high and low activity. |
| 2. | **Decreased cycle time**—Reuse of credit package components on a future deal is now possible. New offerings need not start from scratch, and components of earlier deals are retrievable in a standardized, repeatable fashion. |
| 3. | **Full exposure understood**—System shows all current credit facilities for obligor and all related obligors, allowing current credit decisions to be made with knowledge of full exposure to this customer relationship. |
| 4. | **Auto-generated documents**—Documents are automatically generated by the system based on data entry. Eliminates duplicative data entry and allows company-standard text to be used with data points filled in dynamically as needed. |
| 5. | **Electronic credit file**—All documents related to a credit offering, whether generated automatically by the system or manually uploaded, are stored together with the offering for easy viewing and retrieval. |
| 6. | **Auto-validation of credit offering**—System validations determine if all required fields have been filled in, all required documents have been uploaded, etc., and offering is rejected if incomplete. Auto-validation ensures that once credit management gets the deal, they will have all the information necessary to make a decision. |
| 7. | **Electronic routing and approval**—Credit approval matrix is built into application that can route offering to appropriate approver(s) based on deal parameters such as line of business, deal size, facility type, region and industry, among others. |
| 8. | **Back-office integration**—Facility type or other deal parameters are used to automatically generate booking forms for appropriate back-office processing group(s). Back-office has all the deal terms necessary and can book the deal, or an automatic booking routine can be written for even further efficiency gains. |
| 9. | **Deal archiving**—All data points and documents (even multiple versions of the same document, if desired) are retained and accessible for use in future deals or for satisfying audit and regulatory requirements. |

Making use of these patterns improves performance, eases maintenance, promotes component decoupling and makes the application very adaptable to future changes. These are the hallmarks of a good object-oriented design.

The patterns have been researched extensively, and they are widely adopted in the industry. Furthermore, many of these patterns reflect the way the standard object-oriented patterns are applied to J2EE applications.

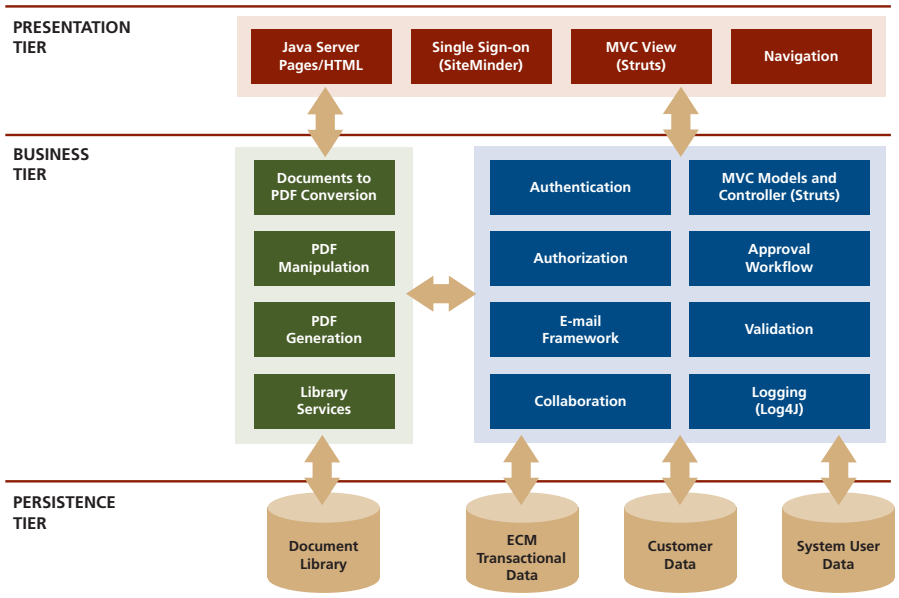
### *A Component-Based View*

The driving force behind the enterprise credit management system software design is a focus on quality factors, such as adaptability, maintainability, extendibility and reusability, through the concept of software components. The system is decomposed into functional components, each of which solves a specific requirement or set of requirements for the application. Figure 7 outlines some of the main components of the enterprise credit management system.

A founding principle of all component-based architectures is the ability for components to be context-independent. Context-independence means that components are easily transferable from their development context to a wide variety of application contexts, and they can be easily replaced by other components with similar functions but different or better qualities.

In the enterprise credit management realm, we gain this context-independence through the use of interfaces and adapters to deliver the functionality of the components. For example, the approval workflow component in Figure 7 will interact the same with other components of the system regardless of the underlying workflow technology product implemented. The same holds true for the library services component; the system behaves the same regardless of the underlying content management implementation. If the institution using the

**Figure 7: Enterprise Credit Management Solution Components**




enterprise credit management solution decides to roll out a new corporate standard for content management, a new adapter to that product simply needs to be implemented to map the library services to that product's application program interface (API).

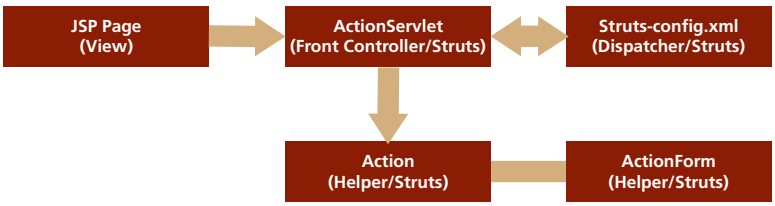
### *Presentation Tier*

The implementation for the presentation tier functionality is the Apache Struts framework, a J2EE-based, open-source implementation of the model-view-controller (MVC) paradigm. The MVC architecture provides a clear separation of the view (the user interface), the model (the data carried by the view) and the controller (the central component that facilitates this process). In addition to these three core components, it is very common to find a helper, which facilitates retrieval of data and data validations. The paradigm has been around since the days of SmallTalk and is a standard for most Web-based systems today.

Apache Struts uses a combination of the front controller, view helper and dispatcher view patterns:

- **Front controller**—The basic idea of this pattern is to route all incoming requests to a central resource, so functions common across many components can be performed here. All incoming requests are routed to the ActionServlet in the Struts framework. The controller manages the handling of the request, including invoking security services, such as authentication and authorization, delegating business processing, managing the choice of an appropriate view, handling errors, and managing the selection of content creation strategies. The centralizing control in the controller reduces business logic in the view and promotes code reuse across requests.
- **View helper**—A view contains formatting code, delegating its processing responsibilities to its helper classes, implemented as JavaBeans or custom tags. Helpers also store the view's intermediate data model and serve as business data adapters. There are multiple strategies for implementing the view component. The Java Server Page (JSP) View Strategy uses JSP as the view component. This is the preferred and most commonly used strategy. All enterprise credit management solution views are implemented as JSP pages that make use of custom tag libraries presupplied by Apache Struts. The helper classes are implemented as FormBeans, which are populated by the incoming data from the form and are responsible for the form-based data validation. The action class can be thought of as the component that forwards control based on logic in coordination with the dispatcher.
- **Dispatcher view**—This pattern solves the problems created by the combination of front controller and view helper patterns in the presentation tier. When these two patterns are employed without the use of a dispatcher, the helper has the data needed by the view and the flow of request handling resides solely with the controller. This amounts to putting entire flow control logic in the controller. If the navigation logic changes, the controller requires code changes, thus reducing the decoupling between the controller and the view. The solution is to introduce a dispatcher in between. The dispatcher can be embedded in the controller or be a separate component. The controller interacts with the dispatcher to determine the navigation flow. Thus changes to navigation logic will not require code changes to the controller; instead, only changes to the dispatcher need to be made. This provides a clear separation between the core processing logic central to all components, such as security, and the navigation logic. Thus the changes are easier to manage. The Struts-config.xml file acts as the dispatcher in Apache Struts. This file contains various

**Figure 8: Presentation Tier**




mappings—for example, FormBeans, action-mappings and forwards.

Figure 8 illustrates the objects described above and their coordination within the presentation tier.

### *Business Tier*

Within the business tier of the application, we find the implementation of the many components that make up the functionality of the enterprise credit management solution. The key object in this layer is the business delegate.

The business delegate acts as a client-side business abstraction; it provides an abstraction for, and thus hides, the implementation of the business services. Using a business delegate reduces the coupling between presentation-tier clients and the system's business services. Depending on the implementation strategy, the business delegate may shield clients from possible volatility in the implementation of the business service API. Potentially, this reduces the number of changes that must be made to the presentation-tier client code when the business service API or its underlying implementation changes.

Figure 9 illustrates the objects used in the business delegate implementation.

The other main pattern used in this tier is the transfer object pattern, which encapsulates business data in a single object. The object is created using a transfer-object-assembler. This serialized object is passed between the presentation and business tiers, after being assembled from data within the persistence tier, and reduces the number of remote calls to the business-service objects. These objects are created on demand and passed by value to the remote client. This allows all calls to the transfer object to be local as opposed to remote.

### *Persistence Tier*

For many applications, persistent storage is implemented with different mechanisms, and there are marked differences in the APIs used to access these different persistent storage mechanisms. Other applications may need to access data that resides on separate systems. For example, the data may reside in mainframe systems, lightweight directory access protocol (LDAP) repositories and others. This scenario is true for the enterprise credit management solution and its data repositories

The enterprise credit management system integrates with its own transactional data store to capture and process data throughout an end-user session with the system. Customer and facility information usually resides within a separate data store, and at most institutions this is a mainframe-based repository. User access, authorization and additional role-based privileges reside within a system user repository; this repository can be an LDAP, Lotus Notes or relational database management system (RDBMS) repository.

We manage access to all of these separate repositories using the data access object (DAO) and factory patterns. The DAO is used to abstract and encapsulate all access to the data source. The DAO manages the connection with the data source to obtain and store data. The business component that relies on the DAO uses the simpler interface exposed by the DAO for its clients. The DAO completely hides the data source implementation details from its clients. Because the interface exposed by the DAO to clients does not change when the underlying data source implementation changes, this pattern allows the DAO to adapt to different storage schemes without affecting its clients or business components. Essentially, the DAO acts as an adapter between the component and the data source. Figure 10 details the DAO objects in the persistence tier.

**Figure 9: Business Delegate Object Model**

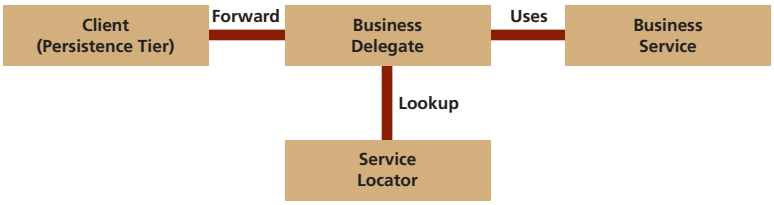
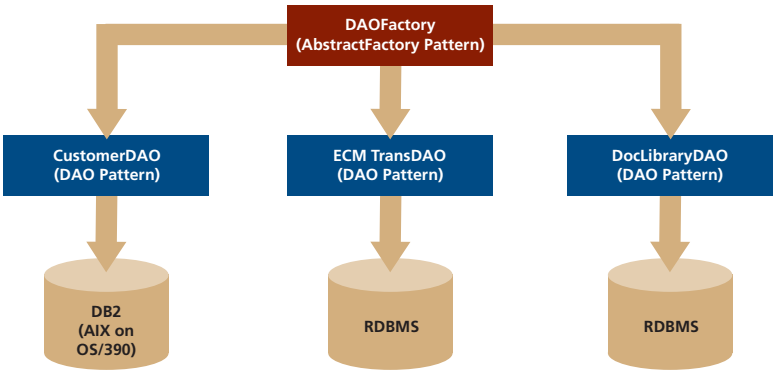


**Figure 10: DAO Object Diagram**




## CONCLUSION

Enterprise credit management is being implemented or contemplated at a number of leading financial institutions either as the natural evolution of their technology platforms or, in some cases, as part of larger initiatives, such as Basel II compliance. The availability today of quality, off-the-shelf components that enable business process workflow and content management make the development of an enterprise credit management solution a very achievable goal. Plus, the business and process efficiency gains make it one solution that few financial institutions can afford to miss out on.

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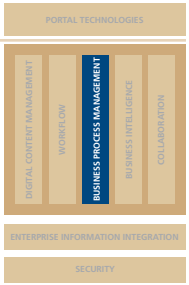
# Business Process Management: Taking All the Right Steps





## Abstract

*As a key component of the digitization stack, business process management technology is touted as a panacea for managing complex workplace processes. However, true business process management (BPM) is more than just implementing a technology solution. This article provides a perspective on how organizations should prepare themselves to ensure that they are ready to develop and implement an effective BPM solution. The authors explain each step in detail and recommend development approaches. They also introduce a variety of custom solutions available to financial services companies.*


## DEJA VUS OR SOMETHING NEW?

Once again the software industry has created a solution—BPM systems—that is positioned to solve many business problems, from capital spending to hiring, across any industry, from defense to retail. Business process management (BPM) can enable cross-functional workflow processes across organizational and geographic boundaries. It can span operating systems and platforms, databases, even network domains, and can include external users such as suppliers and customers.

For those who have been around a while, this all sounds familiar—reminiscent of the buzz about enterprisewide databases, then the mainframe application generators, then enterprise application integration, then Web-based solutions. As capabilities evolved, BPM arose as the new solution, and billions of dollars are being spent on realizing its promise. But as experienced hands know, developing a new technology is only the first step in the solution.

## THE PROMISE OF BPM

BPM solutions automate and manage human interaction with technology systems and business rules, allowing an organization to control complex business integration requirements. With user-friendly graphical tools, the ability to generate specific applications, a standardized framework and compatibility with existing technologies, BPM places rapid development of powerful management tools in hand.

### *Just What Is It?*

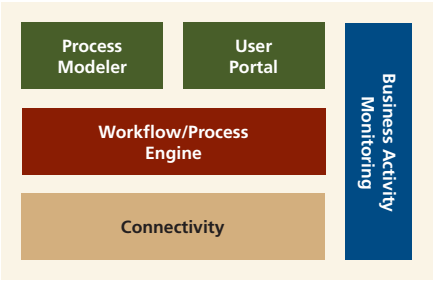
The definition of BPM varies widely and the terms "business process" and "workflow system" are often taken—incorrectly—to mean the same thing. So let's begin by defining the terms.

A business process is a sequence of structured tasks performed in sequence or in parallel by two or more individuals to reach a common goal. A workflow system typically automates the manual tasks in the business process. Thus, workflow is a part of a business process—and a workflow system is a feature of a BPM solution.

The other features are process modeling, connectivity and reporting. Figure 1 depicts a generic architecture for a BPM solution. BPM supports the modeling and manipulation of business processes, along with measurement, tracking and process improvement.

In short, an effective BPM solution is a bridge between "as is" and "should be."

**Figure 1: Basic Architecture for BPM Solutions**



However, no BPM solution can work until the problems to be solved have been completely defined. Managing business processes that span multiple functions—such as portfolio management, risk management, trading and accounting—is not a trivial exercise. In a mature organization, it may be impossible to define business processes in detail, due to the silos that exist within large financial services organizations.

Current-generation BPM solutions support evolutionary approaches, allowing incremental business change. This, of course, implies that effecting the implementation will involve multiple passes to create the necessary requirements and definitions.

## SIX STEPS TO SUCCESS

BPM systems provide a framework for solving business problems by applying rules-based decision making—offering multiple opportunities for financial services firms. BPM solutions can apply to mortgage and loan application processing, dispute management, compliance activities—such as those required by Sarbanes-Oxley and Basel II, for example—insurance risk management, policy processing and many other scenarios.

But implementing a BPM solution is much more demanding than simply hiring a consultant and installing an application. Implementation demands critical prerequisites and specific development approaches. Following are six steps for successful structuring and implementing of BPM projects.

### *Understand Your Process*

The first thing financial services firms must do before implementing a BPM solution is to clearly define their business processes. This may seem obvious, but it is not simple—and it is critical for success. Most BPM software requires a modeling of business processes just to get started, and many firms do not understand their own processes well enough to do that.

Financial institutions must understand that BPM solutions are designed to automate processes, not analyze them. No firm should try to use BPM tools to figure out how the business process should run. That question must be answered before any BPM implementation begins. Existing modeling methodologies—Unified Modeling Language and Business Process Modeling Notation, for example—can be used for this purpose, although their results still must be interpreted to fit a particular BPM solution.

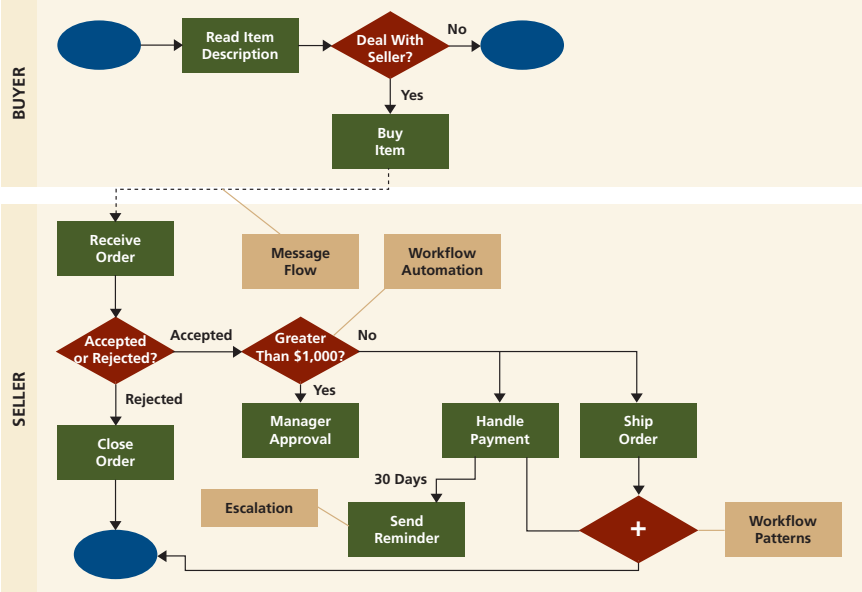
Figure 2 shows how Business Process Modeling Notation is used to depict workflow for a buyer and seller of goods. It allows for the use of workflow design patterns, escalation activity and workflow automation. Many software applications can accept this modeling notation to develop a process flow. In past experience, however, Unified Modeling Language models require significant rework before they are useable in a BPM implementation.

With any off-the-shelf BPM application, a company must configure the solution to fit its needs. This can only happen when it has clearly defined its business process. If a company cannot model its business process, it will end up redesigning its process to fit the BPM solution—a case of the tail wagging the dog.

### *Develop Proof of Concept*

The notion of developing a proof of concept, or prototype, is of major importance in building business applications. It can help organizations understand what the BPM solution's capabilities are. A gap analysis can also reveal the distance between where a process is today and where it needs to be. This lets developers understand how the solution will fit their specific situation, which is helpful since most BPM solutions—like those from PegaSystems and Savvion—

**Figure 2: High-Level Business Process Flow for a Buyer or Seller Using Business Process Modeling Notation**



use proprietary business rules engines that are unfamiliar to end users.

A proof of concept, while extremely useful, takes time to develop, and the company implementing it must be prepared for this. A proof of concept is a worthwhile investment only if a representative subset of a problem can be identified and automated within an acceptable time frame. One successful approach is to allow for a "go, no-go" decision after reviewing the results of the prototype.

Another consideration is the technical complexity of the BPM products. Development life cycles and coding techniques vary greatly. A proof of concept is most feasible when done by developers who are fluent with the chosen BPM tools and a compatible infrastructure is readily available.

### *Leverage Iterative Development*

Adopting an iterative approach to BPM projects, while it may take a bit longer, addresses a number of critical issues associated with successful implementation. BPM solutions represent a major change in the way end users interact with the systems. They must learn to use a process management system that enforces a sequence of tasks with escalation and event triggers.

Iterative development lets users get familiar with the application's interface and functionality as it is being developed, which can facilitate buy-in and training. Users can also provide feedback, which creates a sense of ownership and often gives valuable insights to the development team. Most importantly, this approach can help refine process flow definitions—especially if the business process is unclear in the beginning.

Iterative development relies heavily on careful release management. The project team needs to manage each release closely and balance new functionality with bug fixes.

### *Use Vendor's Database Schema*

Some applications keep data in a normalized relational database; others use an object-oriented data structure that is generated during deployment. In most cases, maintaining the vendor's particular database structure is necessary for calculations and process analysis going forward—a key feature of the BPM solution. Maintaining the database structure streamlines processing and leverages the application framework for maintaining data persistence. Finally, it will simplify implementation by isolating database development from the overall application development.

Circumventing the vendor's schema can introduce design complexity and redundant data occurrences, which make data integrity an issue. The data architecture around objects, which are managed via the BPM workflow, must be considered very carefully. Trade-offs—which conflict with enterprise data standards and practices, yet serve the goals of the business problem—may have to be made. Based on past experience, it is not a good idea to use proprietary components to access data because integration issues may derail overall development.

### *Establish Enterprise Architecture*

Establishing enterprise architecture facilitates development by allowing you to map the BPM framework in terms of services it will leverage and those the enterprise will enable. Most BPM tools contain an array of adaptors, which will combine with common services in most enterprise architectures.

A good example is when user directories are centrally managed with a Lightweight Directory Access Protocol-based tool, which simplifies and consolidates user profiles and access rights. The key factor is that the tool must be extensible so it can handle variances brought out during the requirements phase. This approach allows single sign-on for end users and centralized administration.

### *Evaluate Vendors*

Choose your vendor carefully. Information gathering, developing requirements, writing a request for proposal and vendor selection are all key activities that cannot be shortened or skipped. The BPM solutions industry has a variety of suppliers. Some are pure-play vendors with native business rules engine. Others are enterprise application integration specialists that have enhanced their products. Then there is the subset of vendors that offer industry-specific solutions, not general development tools.

Each type of vendor has its strong points and its areas of best compatibility with your needs. An enterprise application integration vendor is probably the best answer if your business has strong integration requirements. A content management vendor may be best to handle a data-centric solution. And a strong workflow component might call for a pure-play vendor.

This variance among providers makes it essential that companies understand their needs fully and evaluate the candidates completely before choosing a vendor.

## SOLUTION VENDORS: AN ARRAY OF CHOICES

There are a vast range of solutions and vendors that support BPM and here is a brief introduction to a few of these products: PegaSystems, Filenet and Savvion.

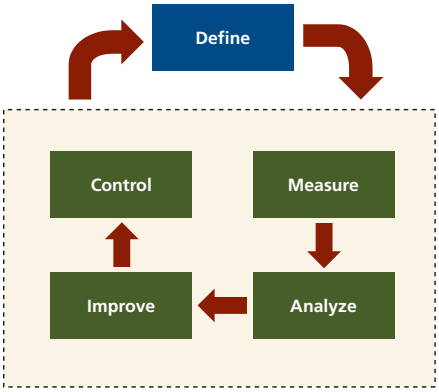
PegaSystems is a pure-play vendor with a strong business rules engine that facilitates workflow. Business rules use

straightforward "If…Then...Else" statements that execute a task based on conditions input by the user. The rules are declarative and may interact with each other. A patented rule resolution algorithm allows the rules engine to identify the appropriate rule to execute at run time.

FileNet is a content-management vendor. Its framework can organize and manage content using workflow functionality to review, approve and distribute structured data, such as documents, and unstructured data, such as images. In addition to well-known imaging functions, the FileNet suite provides a strong connectivity feature that facilitates integration with third-party applications and component services, including business rules engines.

Savvion is a pure-play vendor using a business rules engine with strong monitoring, analysis and reporting functions to monitor business activity. The BPM solution can capture key performance indicators that can be used to measure, analyze and improve the business process. Figure 3 shows the Define, Measure, Analyze, Improve and Control cycle of the Six Sigma process improvement methodology. BPM solutions can be used in the Measure, Analyze, Improve and Control phases. Savvion's solution also has a process simulation capability that allows users to preview the impact of changes to flow, logic, events and rules.

**Figure 3: BPM Solutions Can Be Used in the Measure, Analyze, Improve and Control Phases of the Six Sigma Methodology**




Although each vendor has a comprehensive solution set for BPM implementations, each also has specific areas of strength that may or may not fit a specific enterprise's needs. This only reinforces the need to write a detailed requirements definition so you can map your highest priority requirements to your chosen vendor's natural strengths.

## GETTING PAST THE TECHNOLOGY

BPM solutions comprise several familiar information technology components, but BPM is much more than technology. As this article has pointed out, successful BPM implementation demands clearly defined business processes, a thorough and disciplined development and implementation process, and adequate preparation of the workforce to help the enterprise use the chosen solution effectively.

Implementation is not the end of the story, either. An effective BPM solution allows a company to evaluate and improve its business processes over time to increase overall performance. Through the use of key performance indicators and simulation, BPM solutions permit the fine-tuning of processes on a continuous basis.

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# Document Generation—Digitization in Action

## Abstract

*Paper-based processes are still pervasive in the financial industry today. Undeniably, paper documents provide a tangible, viewable source of information that can be delivered to a customer or used by an employee. Consequently, for digitization to succeed and be widely adopted in the industry, it must offer these same benefits without incurring the traditional cost of paper publication. Moreover, it is imperative that any digitized content be presented in a format that is consistent with printed quality, and a "printable" version must be available upon demand as well. This article discusses a viable approach to digitization that meets all of these requirements.*


## INTRODUCTION

Modern financial services organizations still need publishable documents to support certain processes and meet regulatory requirements. Such documents may not always need to be printed to paper, but they must be self-contained, well structured, formatted and transferable so they can be used by employees or delivered to a customer, government agency or other party.

Many customer-facing and "back-office" functions have been automated for years, employing processes and technologies that produce digital documents. But certain other functions—the cash-flow analysis, underwriting, collateral review and internal credit adjudication activities in commercial loan programs are good examples—have remained largely manual and paper-based. One reason for this is that off-the-shelf software packages have not been available to support all of these processes in an integrated fashion. Such technologies simply could not adapt to the data, process and documentation differences that exist between financial institutions.

However, a new generation of "digitization" capabilities in the financial services industry is changing all of this. Digitization formalizes, through technology, the business processes, content and associated metadata required to serve customers and support good business decisions. One key component of digitization—especially in financial services—is a central source of data that can give employees and other stakeholders access to consistent, accurate and reliable information.

The challenge is providing functional and readable access to this data. Employees of financial services organizations need the ability to retrieve data in a format that they can easily read and comprehend. This is a major driving force behind the need for a process to generate and develop well-formatted documents.

Web applications represent the current, widely used interface for many data presentation needs. The document generation process that we present in this article is not an attempt to replace such applications. On the contrary, the Web page is a well-formatted view that can be the output of this process.

Instead, the process detailed in this article is an alternate method of generating the data that is presented on Web pages and other media. The ability to produce consistently formatted and well-structured documents for desktop computers and to print them at the click of a button greatly enhances the usability of a Web application. The document generation process that we describe herein achieves all of this.

## THE FOUNDATION OF DOCUMENT GENERATION

A successful approach methodology for document generation keeps in mind the key elements of a document generation process and gives a general description of where and what processing needs to occur.

## AT THE HEART OF DIGITIZATION

Data is rarely presentable in its native form. Consequently, the following factors, among others, should be considered when assessing any process for publishing usable, presentable documents:

- The data's location and the format in which it is currently structured
- The ability to separate this current formatting from associated business logic
- The ultimate presentation and formatting of the document, regardless of who the recipient is
- The ability to manipulate the formatting easily

All of these capabilities reside in the realm of digital content management—one of the key areas of the "digitization stack"—a suite of technologies and application domains that captures and manifests the way the institution operates today (see "Digitization: The Time is Here," the lead article in this issue of the *Financial Services Technology Journal*).

### *Key Elements*

The two key elements of the document generation approach are the data source and the template for presentation.

In terms of source, virtually any type will suffice, including relational databases, hierarchical databases, network databases, data stores, documents using extensible markup language (XML), proprietary flat data files, Web-based front-end applications, data mining tools or even manual entry. The only requirement is that the data must exist in a structured format.

The presentation template can be most accurately described as a set of instructions for how to structure the content of the document into the preferred format. This is generally a predefined template, although its format is dependent on the actual document generation approach used.

### *Processing*

The processing that surrounds these key elements is slightly less generalized between various approaches than the elements themselves, but the various approaches have some commonalities. The principal process is applying the template to the data so that the data fills in the appropriate data points in the template, resulting in a presentable document. As easy as this sounds, many other processes need to be in place for this to happen correctly—the topic of the next section.

## A NEW APPROACH TO DOCUMENT GENERATION

The document generation process highlighted in this article can be split into three distinct subprocesses:

- Identifying the content source
- Applying the document template to this content source
- Publishing the result

This process relies on several open source, industry-standard technologies available today: XML, extensible stylesheet language transformation (XSLT) and XSL-formatting objects (XSL-FO), which we describe next.

The salient features of this approach include:

- Modular design, providing the ability to replace individual components.
- Use of open source technologies.
- Flexibility of template.
- Capacity for various input and output formats.

### *Identifying the Content Source*

For purposes of this approach, the content source must be available in XML format. If the source is in an alternate format, most likely a relational database, some processing will need to be done to carry over the data into an XML document.

As an option to simplify this, certain function calls to an Oracle database server, for example, can automatically return data structured as an XML document.

### *Applying the Document Template*

After the data has been formatted in XML, the new approach uses a technology referred to as XSLT—a programming language in which specific style and formatting instructions are applied to an XML document, resulting in a new document composed of both the data from the original XML document and all of the style and formatting information contained in the template.

This output format can be of a few different format types. In this case, the target format is XSL-FO—an open language specification with formatting and style instructions much like hypertext markup language (HTML). However, it is much more specific about formatting and, unlike HTML, is intended for use in multipage documents with distinct line breaks.

### *Publishing the Result*

Once the template has been applied and the evolving document is in XSL-FO format, the next step is to publish the document. Depending on the target format, various tools and processes can be employed to do this. For example, the approach described here results in a portable document format (PDF) document. An open source package called formatting objects processor (FOP) provides this functionality.

A FOP package takes an XSL-FO template as input and outputs a PDF document that is formatted as the XSL-FO template instructs. Other document output formats are currently, or soon will be, available, including rich text format (RTF), scalable vector graphics (SVG), plain text and various print formats, such as printer control language (PCL) and Adobe® PostScript®.

### *Advantages of This Approach*

A key advantage of this approach is its use of open standards. The source, intermediate and output formats are all openly documented standards:

- The World Wide Web Consortium® (W3C) documents XML, XSLT and XSL-FO.
- PDF is an open specification documented by Adobe Systems.

Furthermore, the templates consisting of XSLT and XSL-FO can be altered after the process is in place to change the look and feel of the final output document completely.

### *Disadvantages*

The primary hurdle to implementing this approach is knowledge—knowledge of the template language elements, XSLT and XSL-FO. XSLT is becoming fairly widely used and understood, whereas XSL-FO still has a relatively narrow user base. Even though this is changing, companies will still find it more difficult to locate people with expertise in XSL-FO, in contrast to HTML, which is heavily used and widely known. Nevertheless, someone knowledgeable in XML and HTML would find it easy to learn and master the XSL-FO language elements.

### *Variations on the Approach*

One major variation on this approach might be to change the output format of the template application process. Instead of XSL-FO, the approach could be altered to generate the document directly in HTML, if that is the target format. This approach would have the advantage of skipping the XSL-FO step but the disadvantage of limiting the number of output options. XSL-FO, as previously described, can generate HTML, PDF, SVG and RTF documents. Using HTML for the template application process, however, requires more processing to change to a different output format.

## DOCUMENT MANIPULATION CONSIDERATIONS

There are many considerations for implementing this document generation approach. For example, once a document is generated, it often needs to be modified. Two options are now available for meeting this critical and practical need: interfacing with a document management system and assembling composite documents.

### *Interfacing With a Document Management System*

Any document generation approach needs to consider how it will interface with a document management system, such as FileNet or Documentum. This is important for two reasons:

- Source information, either source XML documents or existing published documents (see the next section), may currently exist in a document management system.
- The output document of this process, or any intermediate document in this process, may need to be placed into the document management system as part of the overall process.

Most document management systems provide application-programming interfaces in Java. However, custom requirements of various applications still might warrant more functionality than that which is made available directly through application program interfaces (APIs). This may necessitate construction of a special interface to these APIs. Building such an interface also has the advantage of lowering the dependence on the vendor document management system.

### *Assembling Composite Documents*

Often, not all required information exists in a single data source. Either some of the information is in other documents existing in a current document management system (see above) or the information exists in an alternate data source.

An approach would need to be developed for converting the existing documents into one of the output formats described in this article and then assembling that into a composite document for final viewing.

Similarly, when multiple documents are being generated from different data sources, again, some assembly process would need to be considered in order to view the multiple documents as a single composite document. For an example of a solution built using such a complex mix of conversion and assembly system in a complete end-to-end digitization solution, see "Enterprise Credit Management: Digitization Comes of Age" in this issue of the *Financial Services Technology Journal*.

*Conclusion*

The document generation approach presented in this article meets the need to produce self-contained, well-formatted documents in the context of a financial services environment. The approach bridges a long-standing gap in the digitization continuum, enabling a financial services organization to create documents for presentation to a customer or to craft well-structured documents for use by employees at any point in a given process, whether that be commercial loans, brokerage services or insurance underwriting. This approach demonstrates the viability of several open industry standards and offers enormous flexibility in the publication process.

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# Enterprise Information Integration: Bridging the Digital Divide

## Abstract

*Integrating access to a company's information has gone from a primitive, technology-driven exercise to a highly sophisticated process that is an absolute requirement for business success. As the need for integration has evolved, so have the tools that make it possible. The latest such tools are known as enterprise information integration (EII) solutions. EII enables an organization to view all its data as though it were housed in a single database, no matter where the information is actually located. This article describes EII solutions and explains how they are different from previous integration solutions. It points out the value a company can realize from an EII implementation. Then, in two appendices, we explain in some detail how EII systems work. We use the term "digital divide" to mean the separation of data that exists in most companies, a major target of current digitization efforts.*


## What is Enterprise Information Integration?

Information integration solutions have been available for years and, until recently, most solutions belong either to the enterprise application integration (EAI) category or the "grandfather" of integration solutions: extract, transform and load (ETL).

EAI provides unrestricted sharing of data and business processes throughout the networked applications and data sources within an organization. ETL combines three database functions into a single tool that allows the movement of information from one database to another.

Enterprise information integration (EII), on the other hand, allows a financial institution to view all back-end information as if it came from one comprehensive database. It does this by using metadata—or underlying data that describes the content, quality, condition and other characteristics of the information in disparate systems—to create a virtual database that gives a single view of data that may be contained in different sources.

An EII solution lets a customer service representative enter a single query and get a response that included information from a Microsoft Excel spreadsheet, a Lotus Notes e-mail, a FileNet image and an Oracle database.

EII can address a number of specific, mission-critical needs, such as providing a single view of customer data without creating a separate customer data repository, maintaining a metadata repository for corporatewide access to customer data or defining logical views of the data.

## EII and the Digital Divide

In the context of this article, we use the term "digital divide" to refer to the difficulty any large company faces in providing consolidated access to its data. Solving the problem, however, is much more complex than that simple definition might imply.

Consider all of the types of digitized data that a financial institution may have today and the tools that its employees can use to access that data. Digitization as we now know it brings together many technologies in a suite—or digitization stack—that captures and manifests the way the institution operates today (see "Digitization: The Time Is Here," the lead article in this issue of the *Financial Services Technology Journal*).

Using these digitization tools can help financial institutions meet the rigors of increasingly competitive markets and growing compliance requirements. But this often means overcoming the digital divide.

### *Can EII Be The Answer?*

In many cases, EII is the ideal bridge for a company's digital divide. Among the capabilities that EII solutions can bring to an organization are:

- Giving customers a consolidated virtual view of data across all accounts and lines of business, increasing the ease and decreasing the complexity of their requests. Customers get easy-to-understand data fields based on industry standards.
- Supporting faster turnaround times for customer requests.
- Using existing technology stack to reduce cost and time of development.
- Providing flexible access to data on a request-response basis.
- Incorporating a visual modeling tool that can be used by business analysts as well as technologists.
- Retrieving data across heterogeneous databases.

### *It Depends On the Question at Hand*

Providing a successful solution remains a challenge, since EII is relatively new in the market. EII vendors face many challenges, including:

- A mix of proprietary database technologies deployed across a company.
- Various versions of structured query language (SQL) used in production.
- Different levels of normalization across heterogeneous databases.
- Demand for request-response messaging to retrieve large amounts of data.
- Inability to repurpose existing data retrieval methods because of non-standard or proprietary coding and processes.
- The need to ensure a high level of scalability and performance without incurring high hardware costs.

From this list of challenges, it is obvious that a systematic approach is essential to decide whether EII is the right solution for your needs. Such an approach will mitigate risk and clearly identify the best solution. The following questionnaire will help you decide whether EII is suitable for your organization. Answer keys and comments, in italics, follow each question.

1. Does your company have multiple platforms and data sources (Oracle, DB2, Extensible Markup Language [XML], Excel, Sybase, etc.)?

   - **Yes** • **No**

*The main problem EII solves is connecting multiple vendor databases installed on different platforms. If your data is predominantly in a common type of data source, you might not need such a tool. If your data is in different data sources, or in one data source but in different schemas, then you should consider an EII tool.*

2. Does your company have a consolidated view of data across the enterprise? More to the point, do your customers have a consolidated view of their data or must they access several reports and manipulate the data they receive?

   - **Consolidated**
   - **Not consolidated**

*If a customer can already get a consolidated view of data across all lines of business within your enterprise, then there is a limited need for an EII tool.*

3. Is your data replicated over a number of data sources?

   - **Yes** • **No**

*An EII tool can provide federated access to all data sources within an enterprise, thus eliminating the need to replicate data.*

4. Select the type of data integration your company would consider feasible in terms of resources, capital and complexity.

   - **Create a consolidated data source by loading data from other databases.**
   - **Access data from the underlying data sources without moving any data**.

*The first choice, loading data into a consolidated source, is not part of an EII solution. The second option describes the basic functionality of EII.*

5. What type of information do you want to provide to your customers, both internal and external?

   - **Request-response**
   - **Publish-subscribe**
   - **Both**

*The core functionality of an EII solution provides information via request-response. Publish-subscribe messaging will probably not be included out of the box and should probably be better supported through a classical EAI approach.*

6. Will your company consider an EII solution for a transaction system; that is, a system that is highly available and loaded?

   - **Yes** • **No**

*The performance of an EII tool is dependent on several factors, so there cannot be a complete guarantee of nonfailure. This variability would not be acceptable in transactional systems so an EII solution may not be ideal.*

7. What is your current turnaround time for creating new reports in response to customer requests?

   - **One to five days**
   - **Two weeks**
   - **One month or more**

*An EII tool should reduce the turnaround time to create reports in response to new customer requests. If you selected the second or third options, you should be able to build a business case for an EII tool.*

## STEP BY STEP

Once you have completed the questionnaire, you will have a fair idea whether an EII solution is worth exploring for your enterprise. As we mentioned earlier, the only effective way to move forward with your evaluation is one step at a time. Recommendations for those three evaluation steps are to define your customers' needs, determine architecture, select a vendor and conduct a performance evaluation.

### *Ask Your Customers*

The first step in evaluating an EII implementation is to identify which requirements are critical for customers. These requirements can be collected and analyzed through a voice of the customer process—a component of the Six Sigma methodology. This is also the stage at which preliminary vendor identification occurs. Different vendors will have different strengths, so their proposed solutions should be examined and understood carefully. One vendor may offer a strong query engine, whereas another could offer a better metadata solution, for example.

### *Architecture Picks the Vendor*

The customer needs that you identify will drive the next step, which is defining the logical architecture of the solution. Each proposed vendor solution should be mapped to the logical architecture to identify which components they will provide and which will be built internally. Architecture scenarios should be developed for each vendor identified in step one, so you can estimate the development effort required. This scenario analysis should lead to a clear vendor choice.

### *Proof Is In the Performance*

To a large extent, implementing EII solutions amounts to the redesign of certain key processes—those associated with the management of information. Best practices associated with BPM apply (see "Business Process Management: Taking All the Right Steps" in this issue of the *Financial Services Technology Journal*). This includes conducting a proof of concept (POC). The POC should be designed to validate your vendor and technology selection. This step is critical to make sure you fully test the capabilities of the tool and understand its limitations. You must define critical success factors and metrics before you begin the POC so you can analyze the outcome and understand what constitutes success. These are some factors to consider in establishing success factors and metrics:

- **Performance**: Optimizing distributed queries over different databases is a complex problem. Rigorous stress testing is essential to help you understand performance limitations.
- **Load balancing**: The ability to offload data extraction and transformation to multiple nodes is the key to scalability. The POC must test whether the tool can run on multiple nodes simultaneously.
- **Reliability**: Distributed queries try to retrieve data from disparate sources, which may result in incomplete results. The POC should determine whether the tool is able to return a consistent result each time a same query is executed.
- **Flexibility/extensibility**: The EII tool will most likely need to be extended if it is to provide all the functionality defined in the logical architecture. The POC should test all feasible extensibility scenarios.
- **Quality of data**: There is a limit to the amount of on-the-fly transformations that can be performed on the data. Stress testing will help you understand the impact of such transformations on performance.
- **Security**: The EII tool will have to integrate with the company's access and entitlement systems, so the POC should test its security integration capabilities.

Once the POC is completed, you can organize the implementation phase, which is most effectively done in a series of plateaus with a number of cutout options.

## CONCLUSIONS

We have tried to explain what is involved in implementing an EII solution, from deciding whether EII is right for you through POC. It is important to state, however, that the EII market is still emerging and the technology is not mature. As yet, there are only a few EII production implementations in the financial services industry. At the same time, we note that several such implementations are in the pipeline, indicating a growing acceptance of EII as a solution.

## APPENDIX A: CORE FUNCTIONALITIES

This appendix demonstrates the typical functionalities of an EII solution. Diagram 1 presents the various layers of an EII tool, followed by explanations of the functionality of each layer and a glossary of acronyms.

### *Client Interfaces*

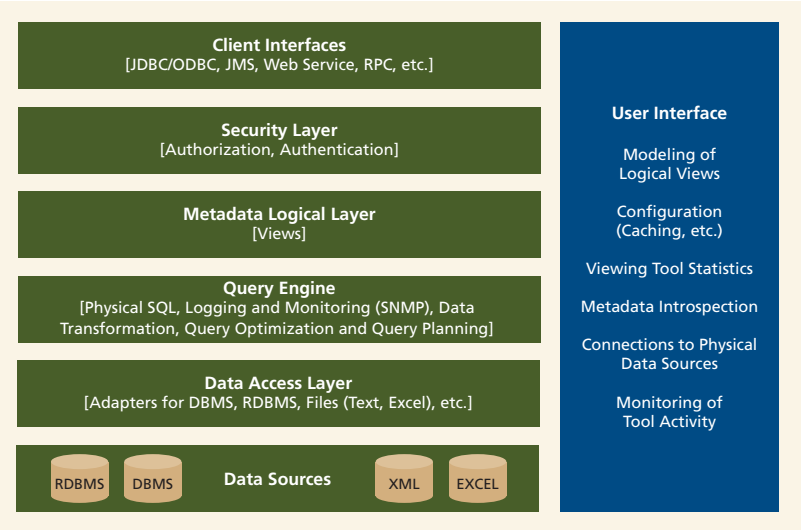
An entry point for external users that will allow them to view the data contained in the underlying data sources.

- Java Database Connectivity/Open Database Connectivity (JDBC/ODBC) provides external systems with a JDBC or ODBC interface to use the EII tool.
- Java Messaging Services (JMS) lets an external management and monitoring system integrate with the EII tool through workflow logic.
- Web service provides access to data exposed as a Web Service via Simple Object Access Protocol (SOAP).
- Remote Procedure Calling (RPC) provides an application program interface (API) for system-to-system integration.

### *Security Layer*

- **Authorization**:
  - Checks entitlements at the data level and function level.
  - Imports entitlements from a third-party system.
  - Imports from directories such as Lightweight Directory Access Protocol (LDAP).
- **Authentication**:
  - Uses token-based authentication for system-to-system integration.
  - Authenticates at both the server and client levels.

**Diagram 1: Core Functionality**




### *Metadata Logical Layer*

Views form a virtual layer over underlying data sources to provide a consolidated, logical view. These views contain fields that map to disparate data sources.

### *Query Engine*

- Physical SQL transforms a logical request for data into SQL statements distributed over several databases.
- Logging and monitoring tool should support Simple Network Management Protocol (SNMP) for the integration with monitoring and management systems.
- Data transformation is provided at the database function level (SUBSTR, CONCAT, etc.) and response format level (XML, comma-separated values [CSV], etc.). Currently the transformation into several response formats is limited.
- Query optimization, a key function of EII tools, generates dynamic SQL that is optimized to go across multiple databases installed on different systems.
- Query planning diagrams the path a query will take at run time. This helps guide design decisions.

### *Data Access Layer*

- EAI technologies have made the availability of mature and cost-efficient adapters widespread.
- Most EII vendors will use third-party adapters or bundle their own.

**Data Sources** are currently outside a typical EII solution.

### *User Interface*

The interface should provide the following functions:

- Modeling of logical views
- Configuration (caching, etc.)
- Monitoring of tool statistics
- Metadata introspection
- Connection to physical data sources
- Monitoring of tool activity

*Note:*
*The above functions are representative only.*

## APPENDIX B: ADDITIONAL FUNCTIONALITIES

EII vendors realize that their solutions must go beyond being a query engine powered by metadata. They know they must provide additional functionalities. Diagram 2 represents the diagram from Appendix A with some possible extended functions highlighted in gold.

### *Security Layer*

- **Data encryption**: Some vendors recommend encrypting the entire data payload, others recommend only encrypting parts of a document.
- **Web service security**: Support of a Web service security mechanism can help developers of Web services secure SOAP message exchanges.

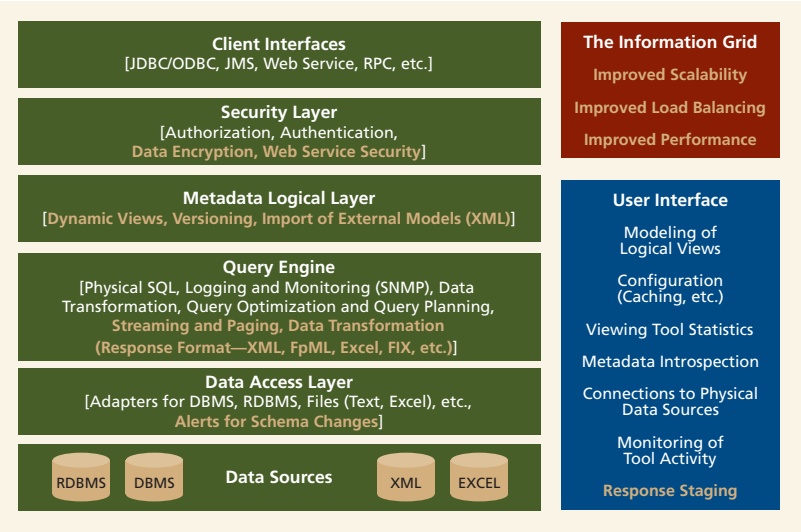
### *Metadata Logical Layer*

- **Dynamic views**: A customer at run time should be able to select the fields that he or she wants to retrieve.
- **Versioning**: With XML support being an important part of the EII solution, versioning of XML schemas becomes very important. Each customer will need a customized schema and the EII tool should be able to maintain up-to-date versions.
- **Import of external models**: The EII tool should be able to represent views in a canonical form so that external models can be imported successfully.

### *Query Engine*

- Streaming and paging mechanisms can help make dramatic improvements to the performance of an EII tool.
- Data transformation capabilities need to be included in an EII toolset because of the increasing need to comply with industry standards.

**Diagram 2: Additional Functionality**




### *Data Access Layer*

- **Alerts for schema changes**: Since EII tools can retrieve data from any information source, there is a risk of a schema change that results in failure. Support of proactive alerts can be important.

### *User Interface*

- **Response staging**: While request-response messaging is the preferred format of an EII system, it might not always be feasible due to the large amount of data to be retrieved for a specific query. In these cases, the tool should be able to stage a response in a central repository for later retrieval.
- **Response scheduling**: Providing scheduled execution of requests can be very useful.

### *The Information Grid*

- Grid computing introduces a new concept to information technology infrastructures because it supports distributed computing over a network of heterogeneous resources and is enabled by open standards.

Improvements will be found in:

  – Scalability
  – Load balancing
  – Performance

- An information grid optimizes under-used resources, lowers capital expenditures and reduces the total cost of ownership. This addition to the EII solution extends beyond data processing and into information management.

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# Digitization and Security in Financial Services

## Abstract

*Digitization—in its simplest form, converting paper documents to digital files—has created a wealth of opportunity for financial services firms, providing enormous efficiency improvements and greatly enhanced customer service. It also has created a tremendous level of concern over security of the information once it is in digital form. That concern touches companies, their customers and their regulators. It has also impelled state and federal legislators to pass laws demanding that organizations ensure the privacy, confidentiality and integrity of digitized information. This article discusses the background and validity of those concerns, why these concerns are critical to financial services firms and how these firms can reduce the risk for their information, their customers and themselves.*


## Digitized Information: Today's Double-Edged Sword

Digitization—in its simplest form, the process of converting information from paper to digital form—facilitates data access, compliance and analytical insight. It makes most internal processes more efficient, from audit to customer service. Indeed, digitization has transformed many traditional "brick and mortar" companies into true e-businesses.

At the same time, however, companies' digital information assets represent an attractive target for security compromises that can lead to abuse, misuse and theft. Even as these abuses grow, regulators and lawmakers are imposing new compliance standards focused on data privacy, confidentiality, integrity and availability. These concerns and requirements must be addressed by the disciplines of information security.

Information security is being transformed from a technology-centered management practice—focusing on security of the system through firewalls, proxies and other perimeter defenses—to a core business discipline that deals with security of the information the system holds. As part of that transformation, privacy, confidentiality, integrity and availability of information have created major risk management, regulatory and compliance requirement issues.

These requirements apply to all industry sectors but are especially important in financial services. This is evidenced by the fact that major legislation dealing with financial controls—including the Sarbanes-Oxley Act, Gramm-Leach-Bliley Act, Basel II Capital Requirements Accord and USA PATRIOT Act's Know Your Customer provision—stipulate data privacy and integrity as key tenets. They all require that a financial institution maintain a security policy approved by its board of directors.

Naturally, digitization expands the scope of these requirements. Today, digitized financial information, customer information and related documents—such as cleared checks, account applications, financial instruments and written trade instructions—are a primary security concern in the financial services industry.

As a final point, access to confidential information, identity theft or unauthorized access to institutional fund transfers can be extremely attractive to criminals—and costly to financial institutions.

All of these factors have combined to put financial institutions at the forefront of efforts to secure digitized content.

## Computer-Based Theft: A Big and Growing Business

The number of Internet security incidents reported to authorities is growing at an alarming rate. Identity theft, for example, is one of the fastest growing white-collar crimes. The Federal Trade Commission reports that the number of these complaints rose from 86,323 in 2001 to 214,905 in 2003.

Equally staggering is the cost of these crimes—identity theft in 2002 cost individuals $5 billion, whereas financial institutions lost about $48 billion.

At the same time, digital criminals are getting smarter, faster and harder to catch. Operating system security experts report that attackers now routinely reverse-engineer operating system security patches in order to design new attacks. As a result, the time between the release of a patch and the creation of new attack tools has dropped from months to days. Financial institutions are among the favorite targets for intruders.

Recent database breaches at large financial institutions received significant media coverage and contributed to heightened awareness—and concern—over the potential vulnerability of confidential data.

State and federal lawmakers reacted with legislation trying to institutionalize the integrity, confidentiality and availability of key financial data. The legislation applies to all financial institutions with particular impact in three key areas—privacy, confidentiality and integrity.

- **Privacy**—Organizations that acquire, maintain and share customer information are now subject to financial, operational and compliance risks associated with enrolling customers and providing products and services to them. Examples of such risks include violation of customer privacy preferences.

  In addition, regulations such as Sarbanes-Oxley, Basel II, the USA PATRIOT Act's Know Your Customer, Gramm-Leach-Bliley Act and California Senate Bill 1386 require stringent security measures and impose tight deadlines for implementation. Companies today are being forced to put more robust infrastructures in place to mitigate the risks associated with a security breach.
- **Confidentiality**—In January 2003, the Federal Financial Institutions Examination Council issued new guidance that went beyond the requirements of Gramm-Leach-Bliley. The new guidance requires financial institutions to protect all information assets, not just customer information. The council issued recommendations for specific security processes that financial institutions must put in place to stay compliant with the expanded requirements.
- **Integrity**—Legislation requires companies to implement information management controls that can assure the integrity and availability of key financial information.

The list of regulations and legislation is long and growing, intensifying the pressure on financial institutions to

## A PARTIAL LIST OF LEGISLATION REQUIRING FINANCIAL INSTITUTIONS TO SECURE FINANCIAL INFORMATION

Gramm-Leach-Bliley Act (1999)—Protects consumers' personal financial information held by financial institutions

Sarbanes-Oxley Act (2002)—Places strict requirements on company Boards and Officers to proactively prevent mishandling of information

California SB 1386 (the California Data Security Act)—Requires control of privacy of individual financial information

New York Reg. 173—Mandates the active encryption of sensitive financial information sent over the Internet

Homeland Security Information Sharing Act (HSISA, H.R. 4598)—Outlines security rules and regulations

Healthcare Insurance Portability and Accessibility Act (HIPAA)—Places liability on anyone who fails to properly protect patient health information, including bills and health-related financial information

International Organization for Standardization (ISO) 17799—Defines an extensive approach to information security, including communications systems requirements for information handling and risk reduction

Basel II Capital Requirements Accord

European Union Data Protection Directive—Mandates protection of personal data

Japanese Protection for Personal Information Act, Kojin Joho Hogo HouA (May 2003)

U.S. Federal Trade Commission, 16 CFR Part 314

U.S. Security Exchange Commission (SEC) Final Rule, Privacy of Consumer Financial Information (Regulation S-P), 17 CFR Part 248 RIN 3235-AH90 and 17 CFR Part 210's records retention

International Accounting Standards Board's accounting procedures from American Institute of Certified Public Accountants and Financial Accounting Standards Board

SEC, National Association of Security Dealers (NASD), FDA 21 CFR Part 11

NASD rules 2711 and 3010

New Jersey AB 2553—A new bill under consideration that prohibits companies from displaying and using social security numbers

implement robust, effective security systems for their digital information (see "Partial List of Legislation Requiring Financial Institutions to Secure Financial Information").

Considering the breadth, nondiscretionary nature and the aggressive deadlines of these and emerging rules, financial institutions now must consider an information security framework as part of the overall risk management and compliance framework. Obviously, there is a clear need to develop and implement information security controls that can protect an enterprise and its key information assets and help the organization comply with relevant regulations as rapidly as possible.

## FINDING THE RIGHT SOLUTION

For simplicity, we will confine the discussion of organizational security implementation to an enterprise-centric point of view, meaning all the relevant information assets reside within the enterprise. Other, more complex information management models exist, of course, but they are beyond the length and scope of this article.

First, it is important to understand the scope of digitization within financial institutions today. Digitization as we know it today brings together many technologies in a suite—or digitization stack—that captures and manifests the way the institution operates today (see "Digitization: The Time is Here," the lead article in this issue of the *Financial Services Technology Journal*).

Using these digitization technologies can help financial institutions meet the rigors of increasingly competitive markets and growing compliance requirements. The security domains in an organization that employs these tools would be organized into layers that protect the entire enterprise from the outside in. These domains could be described as "security of exclusion" and "security of inclusion"—layers that would act in concert to protect the enterprise and its assets.

The more traditional security of exclusion is focused on the network, perimeter and platform, intending to deny access to the enterprise from the outside. The newer security of inclusion is based on knowing and managing the identity of users who request access to the enterprise resources. This allows protection of applications and data, as well as control of users. The business processes and technology that enable the security of application, data and user information are known as identity management. This knowledge set is at the heart of the overall security framework and makes up the fastest-growing area of expertise in the information security field.

The disciplines involved in identity management include traditional security technologies, such as encryption, digital signatures, intrusion detection and prevention, nonrepudiation and virus protection, as well as still-maturing technologies, such as authentication, authorization, entitlements, administration and provisioning (see "Security Definitions").

## FINANCIAL SERVICES: AT THE FRONT LINES

Many financial services firms are just now awakening to the increased security demands that result from digitization. Consider a bank that made a business decision to move to a paperless office. Its first two steps were eliminating the storage of paper-based mortgage applications and delivering electronic—digitized—images of cleared checks to its customers.

Cleared checks and mortgage applications typically contain private, personal information about customers. Therefore, the newly converted digitized information is immediately subject to all regulations dealing with privacy and confidentiality. In addition, the integrity of this information needs to be assured so that any legal dispute can be resolved.

It is tempting to think that digitized information can be protected using familiar technologies, such as encryption and digital signature. Indeed, encryption is designed to protect data confidentiality and digital signatures protect message integrity.

Unfortunately, this approach is incomplete. Encrypting a digitized image may make it difficult to search, for example. Also, digitized information may have to be archived for long periods of time—seven years in the case of SEC-regulated documents. Retrieving encrypted documents requires the use of an encryption key, which may have expired by the time the document is needed.

## WHAT TO DO: THE SHORT LIST

The list of concerns is lengthy, and we have only touched on a representative sample of them. An appropriate solution needs to take into account all possible uses and potential vulnerabilities of the digitized information. A well thought-out information security strategy, coupled with an end-to-end information security framework, can address most security concerns for the digitized organization, including the following:

- Protection of digitized content
  - Creating and maintaining the data control repository so the organization knows where information is and who has access to it

## SECURITY DEFINITIONS

**Authentication**—Verifying that the identity of an entity is exactly what the entity (e.g., a user) claims it to be. Authentication can be done via the use of user ID and passwords, security tokens, smart cards or biometrics such as a fingerprint or iris scan. In general, authentication is said to be stronger when a user seeking access can provide at least two of the following three items: something you know, something you have, something you are.

**Entitlements**—Determining what information and computing resources the identified party is allowed to access. Entitlements are enforced through an authorization mechanism that automatically enforces a management policy regarding the use of the resource. This policy could be role-based, rules-based or a combination of the two. Clearly, authorization is driven by and depends on reliable authentication.

**Confidentiality**—Defining the rules and processes that can protect certain information from unauthorized use. Often, confidentiality is enabled through encryption. Confidentiality is stronger if the organization also implements access controls based on authentication, authorization and entitlements.

**Integrity**—Ensuring that data in a file or a message traversing the network remains unchanged or that any data received matches exactly what was sent. Data integrity deals with the prevention of accidental or malicious changes to data or message content.

**Auditing and Accountability**—Collecting and analyzing data so administrators and others, such as information technology auditors, can verify that authentication and authorization rules are producing the intended results. Individual accountability for attempts to violate the intended policy depends on monitoring relevant security events, which should be stored securely and time-stamped using a trusted time source in a reliable log of events, also known as an audit trail or a chain of evidence archive. This audit log can be analyzed to detect attempted or successful security violations. The monitoring process can be implemented as a continuous automatic function, as a periodic check or as an occasional verification that proper procedures are being followed. The audit trail may be used by security administrators, internal audit personnel, external auditors and government regulatory officials, as well as in legal proceedings.

- Access control to digitized content
  - Strong authentication of identities
  - User roles, authorization, permissions, entitlements and policies
  - Access by authors, publishers and readers
- Application/business logic
  - Maintaining the ability to search the digitized content
  - Managing and preserving encryption keys for future retrieval
  - Providing accountability for all actions taken with the digitized content

## MOVING TO A SOLUTION

In the past, digitization and security were seen as concerns for an organization's technology experts. Today, we know better. They are critical business processes that deserve attention and direction from the highest levels of an organization.

For financial institutions, financial information is an asset that is every bit as important to its operations as a fleet of trucks would be to a shipping company. It deserves the same level of protection from theft and misuse.

As part of the encompassing digitization framework that a financial institution chooses, a comprehensive, end-to-end security framework can help protect information privacy, confidentiality and integrity while letting the organization continue to apply its current business processes to newly digitized information.

## ABOUT THE AUTHOR

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